

4.3 Let ν be the counting measure on $\mathbb{I}_{0,1}$ and $\int_{\mathbb{I}_{0,1}} h d\nu < \infty$ for some non-negative function h . Show that $B = \{x = h(x) > 0\}$ is countable.

Let $B_n = \{x = h(x) \geq \frac{1}{n}\} \Rightarrow \cup B_n = B$, $B_{n+1} \supset B_n$, $B_n \rightarrow B$ as $n \rightarrow \infty$

For each n , h is non-negative $\Rightarrow \nu(B_n)/n \leq \int_{B_n} h d\nu \leq \int_{\mathbb{I}_{0,1}} h d\nu < \infty$.

$\Rightarrow B_n$ is a finite set for every $n \in \mathbb{N}$. and $B = \cup B_n$ is countable.

4.4 Prove Proposition 4.1.2

Let $\nu, \mu, \mu_1, \mu_2, \mu_3, \dots$ be σ -finite measures on measurable space $(\mathcal{D}, \mathcal{F})$

(i) If $\mu_1 \ll \mu_2$ and $\mu_2 \ll \mu_3$, then $\mu_1 \ll \mu_3$ and $\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3}$ a.e. (μ_3)

(ii) Suppose that μ_1 and μ_2 are dominated by μ_3 . Then for any $\alpha, \beta \geq 0$, $\alpha\mu_1 + \beta\mu_2$ is dominated by μ_3 and

$$\frac{d(\alpha\mu_1 + \beta\mu_2)}{d\mu_3} = \alpha \frac{d\mu_1}{d\mu_3} + \beta \frac{d\mu_2}{d\mu_3} \text{ a.e. } (\mu_3)$$

(iii) If $\mu \ll \nu$ and $\frac{d\mu}{d\nu} > 0$ a.e. (ν) then $\nu \ll \mu$ and $\frac{d\nu}{d\mu} = \left(\frac{d\mu}{d\nu}\right)^{-1}$ a.e. (μ)

(iv) Let $\{\mu_n\}_{n=1}^\infty$ be a sequence of measures and $\{\alpha_n\}_{n=1}^\infty$ be a sequence of positive real numbers, i.e. $\alpha_n > 0$ for all $n \geq 1$.

Define $\mu = \sum_{n=1}^\infty \alpha_n \mu_n$.

(a) $\mu \ll \nu$ iff $\mu_n \ll \nu$ for each $n \geq 1$ and in this case

$$\frac{d\mu}{d\nu} = \sum_{n=1}^\infty \alpha_n \frac{d\mu_n}{d\nu} \text{ a.e. } (\nu)$$

(b) $\mu \perp \nu$ iff $\mu_n \perp \nu$ for all $n \geq 1$

(1) Since $\mu_1 \ll \mu_2 \ll \mu_3$, $\mu_3(A) = 0 \Rightarrow \mu_2(A) = 0 \Rightarrow \mu_1(A) = 0 \Rightarrow \mu \ll \mu_3$

For any σ -finite measures μ and ν on $(\mathcal{D}, \mathcal{F})$

suppose $\mu \ll \nu$, and for any non-negative function h , we have

$$\int_{\mathcal{D}} h d\mu = \int_{\mathcal{D}} h \frac{d\mu}{d\nu} d\nu.$$

For any $A \in \mathcal{F}$ ($A \subset \mathcal{D}$) let $h = \mathbb{I}_A \frac{d\mu_1}{d\mu_2}$, non-negative.

$$\Rightarrow \mu_1(A) = \int_A d\mu_1 = \int_{\mathcal{D}} \mathbb{I}_A \frac{d\mu_1}{d\mu_2} d\mu_2 = \int_{\mathcal{D}} h d\mu_2 = \int_{\mathcal{D}} h \frac{d\mu_2}{d\mu_3} d\mu_3 = \int_{\mathcal{D}} \mathbb{I}_A \frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3} d\mu_3 = \int_A \frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3} d\mu_3$$

$$\Rightarrow \frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3} \quad \text{a.e. } (\mu_3)$$

(i') $\alpha\mu_1 + \beta\mu_2$ is a valid σ -finite measure on $(\mathcal{X}, \mathcal{F})$ since $\alpha > 0, \beta > 0$.

$$\forall A \in \mathcal{F} \quad (A \subset \mathcal{X})$$

$$\begin{aligned} (\alpha\mu_1 + \beta\mu_2)(A) &= \alpha\mu_1(A) + \beta\mu_2(A) = \alpha \int_A \frac{d\mu_1}{d\mu_3} d\mu_3 + \beta \int_A \frac{d\mu_2}{d\mu_3} d\mu_3 \\ &= \int_A \left(\alpha \frac{d\mu_1}{d\mu_3} + \beta \frac{d\mu_2}{d\mu_3} \right) d\mu_3 \end{aligned}$$

Since $\alpha \frac{d\mu_1}{d\mu_3} + \beta \frac{d\mu_2}{d\mu_3}$ is nonnegative and measurable.

$\alpha\mu_1 + \beta\mu_2 \ll \mu_3$ and its RN derivative w.r.t. μ_3 is $\alpha \frac{d\mu_1}{d\mu_3} + \beta \frac{d\mu_2}{d\mu_3}$.

(ii') $\forall A \in \mathcal{F}$. Let $h = I_A \cdot \left(\frac{d\mu}{d\nu} \right)^{-1}$

$$\nu(A) = \int_{\mathcal{X}} I_A d\nu = \int_{\mathcal{X}} I_A \left(\frac{d\mu}{d\nu} \right)^{-1} \left(\frac{d\mu}{d\nu} \right) d\nu = \int_{\mathcal{X}} h \frac{d\mu}{d\nu} d\nu = \int_{\mathcal{X}} h d\mu = \int_A \left(\frac{d\mu}{d\nu} \right)^{-1} d\mu.$$

$\Rightarrow \nu \ll \mu$. and ν 's RN derivative w.r.t. μ is $\left(\frac{d\mu}{d\nu} \right)^{-1} = \frac{d\nu}{d\mu}$ a.e. (μ) .

(iv) sum of nonnegative sequence is 0 iff. every term in the sum is 0.

$\Rightarrow \mu \ll \nu \Leftrightarrow \mu_n \ll \nu$ for all $n \geq 1$.

for any $A \in \mathcal{F}$, $A \subset \mathcal{X}$. By corollary 2.3.1.

$$\mu(A) = \sum_{n=1}^{\infty} \int_A \alpha_n \frac{d\mu_n}{d\nu} d\nu = \int_A \left(\sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu} \right) d\nu.$$

$$\Rightarrow \frac{d\mu}{d\nu} = \sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu} \quad \text{a.e. } (\nu).$$

α_n 's are positive real numbers $\Rightarrow \mu(B) = 0 \Rightarrow \mu_n(B) = 0$ for all $n \geq 1$

$\mu_n(B_n) = 0$ and $D = \cap B_n \Rightarrow \mu_n(D) = 0 \Rightarrow \mu(D) = 0$

$$\nu(D^c) = \nu(\cup B_n^c) \leq \sum \nu(B_n^c) = 0$$

$\Rightarrow \nu \perp \mu$. iff $\mu_n \perp \mu$ for all $n \geq 1$

4.5. Let Lebesgue measure restricted to $\mathbb{R}^+$, denoted by m_1 i.e. $m_1(A) = m_1(A \cap \mathbb{R}^+)$
 Let counting measure restricted to $\mathbb{N}$, denoted by λ_1 i.e. $\lambda_1(A) = \lambda_1(A \cap \mathbb{N})$

(a) $\mu = \mathcal{N}(0,1)$ $\nu = \text{Exp}(1)$

Define $\mu_a(A) = \mu(A \cap \mathbb{R}^+)$ $\forall A \in \mathcal{B}(\mathbb{R})$, then $\mu_a \ll m_1$. μ_a is the a.e. part of μ w.r.t. ν .

suppose $B \in \mathbb{R}^+$, s.t. $(\mu - \mu_a)(B) = 0$ and $\nu(B^c) = 0$.

$\nu \ll m_1$ and $\frac{d\nu}{dm_1} = e^{-x} > 0$ a.e. (m_1) By proposition 4.1.2. $\mu_a \ll m_1 \ll \nu$.

and $\frac{d\mu_a}{d\nu} = \frac{d\mu_a}{dm_1} \left(\frac{d\nu}{dm_1} \right)^{-1} = \frac{1}{\sqrt{2\pi}} e^{-x^2/2+x}$ a.e. (ν)

(b) $\mu = \text{Exp}(1)$ $\nu = \mathcal{N}(0,1)$

$\frac{d\nu}{dm} > 0$ a.e. (m) $m \ll m_1 \Rightarrow \mu \ll m_1 \ll m \ll \nu$.

Lebesgue decomp. is unique $\Rightarrow \mu$ is a.e. part of μ with respect to ν .

and $\frac{d\mu}{d\nu} = \frac{d\mu}{dm} \left(\frac{d\nu}{dm} \right)^{-1} = \sqrt{2\pi} e^{-x+x^2/2} \mathbb{I}\{x > 0\}$ a.e. (ν)

(c) $\mu = \mu_1 + \mu_2$ $\mu_1 = \mathcal{N}(0,1)$ $\mu_2 = \text{Poisson}(1)$ $\nu = \text{Cauchy}(0,1)$

Cauchy density is strictly positive on $\mathbb{R}$, $\Rightarrow \mu_1 \ll m \leq \nu$.

$\nu(\{0\} \cup \mathbb{N}) = 0$ $\mu_2(\{0\} \cup \mathbb{N}) = 0$ says that $\mu_2 \perp \nu$.

So μ_1 is the a.e. part of $\mu_1 + \mu_2$ w.r.t. ν .

and $\frac{d\mu_1}{d\nu} = \frac{d\mu_1}{dm} \left(\frac{d\nu}{dm} \right)^{-1} = \sqrt{\frac{2}{\pi}} (1+x^2) e^{-x^2/2}$ a.e. (ν)

(d) $\mu = \mu_1 + \mu_2$. $\nu = \text{Geometric}(p)$, $0 < p < 1$, where $\mu_1 = \mathcal{N}(0,1)$ $\mu_2 = \text{Poisson}(1)$

$\mu_2(A) = \mu_2(A \cap \{0\})$ $\forall A \in \mathcal{B}(\mathbb{R}) \Rightarrow \mu_{2a}(A) = \mu_2(A \cap \mathbb{N})$ $\forall A \in \mathcal{B}(\mathbb{R})$

Since $(\mu_1 + \mu_2)(\mathbb{N}) = 0$ $\nu(\mathbb{N}^c) = 0$

and $\nu(A) = 0 \Rightarrow A \cap \mathbb{N} = \emptyset \Rightarrow \mu_{2a}(A) = 0$,

$\mu_{2a} \ll \nu$ and $(\mu_1 + \mu_2) \perp \nu$

So μ_a is the a.c. part of $(\mu_1 + \mu_2)$ w.r.t. ν and $\mu_a \ll \lambda_1 \ll \nu$ where λ_1 is counting measure.

$$\Rightarrow \frac{d\mu_a}{d\nu}(w) = \frac{d\mu_a}{d\lambda_1} \frac{d\lambda_1}{d\nu} = \left(\sum_{w \in \mathcal{N}} I_{\{w=k\}} (1-p)^k p \right)^+ \left(\sum_{w \in \mathcal{N}} I_{\{w=k\}} \frac{e^t}{k!} \right)$$

$$= \sum_{w \in \mathcal{N}} \frac{(1-p)^{r_k}}{e p^{k-1}} I_{\{w=k\}} \text{ a.e. } (\nu)$$

(e) $\mu = \mu_1 + \mu_2$, $\nu = \nu_1 + \nu_2$, $\mu_1 = N(0,1)$, $\mu_2 = \text{Poisson}(1)$, $\nu_1 = \text{Cauchy}(0,1)$, $\nu_2 = \text{Geometric}(p)$, $0 < p < 1$

$$\mu_{\nu_1}(A) = \mu_1(A \cap \{0\}), \mu_{\nu_2}(A) = \mu_2(A \cap \mathcal{N}) \quad \forall A \in \mathcal{B}(\mathbb{R})$$

$$\mu_{\nu_1}(\mathbb{R} \setminus \{0\}) = 0, (\nu_1 + \nu_2)(\{0\}) = 0 \Rightarrow \mu_{\nu_1} \perp \nu_1 + \nu_2$$

$$\mu_1 \ll \nu_1, \mu_a \ll \nu_2 \Rightarrow \mu_1 + \mu_a \ll \nu_1 + \nu_2.$$

$\Rightarrow \mu_1 + \mu_a$ is absolute continuous part of $\mu_1 + \mu_2$ w.r.t. $\nu_1 + \nu_2$

$\mu_1 \perp \nu_2, \mu_a \perp \nu_1, \mu_1 \ll \nu_1, \mu_a \ll \nu_2$ for any $A \in \mathcal{N}$.

$$\begin{aligned} (\mu_1 + \mu_a)(A) &= \mu_1(A) + \mu_a(A) \\ &= \int_A \frac{d\mu_1}{d\nu_1} d\nu_1 + \int_A \frac{d\mu_a}{d\nu_2} d\nu_2 \\ &= \int_A \frac{d\mu_1}{d\nu_1} I_{\mathbb{N}^c} d\nu_1 + 0 + \int_A \frac{d\mu_a}{d\nu_2} I_{\mathcal{N}} d\nu_2 + 0 \\ &= \int_A \frac{d\mu_1}{d\nu_1} I_{\mathbb{N}^c} d\nu_1 + \int_A \frac{d\mu_1}{d\nu_1} I_{\mathbb{N}^c} d\nu_2 + \int_A \frac{d\mu_a}{d\nu_2} I_{\mathcal{N}} d\nu_2 + \int_A \frac{d\mu_a}{d\nu_2} I_{\mathcal{N}} d\nu_1 \\ &= \int_A \frac{d\mu_1}{d\nu_1} I_{\mathbb{N}^c} d(\nu_1 + \nu_2) + \int_A \frac{d\mu_a}{d\nu_2} I_{\mathcal{N}} d(\nu_1 + \nu_2) \\ \Rightarrow \frac{d(\mu_1 + \mu_a)}{d(\nu_1 + \nu_2)} &= \frac{d\mu_1}{d\nu_1} I_{\mathbb{N}^c} + \frac{d\mu_a}{d\nu_2} I_{\mathcal{N}} \\ &= \sqrt{\frac{2}{\pi}} (1+x^2)^{-1} e^{-x^2/2} I_{\{x \in \mathbb{R}\}} + \sum_{k \in \mathcal{N}} \frac{1}{e p^{(1-p)^{k-1}} k!} I_{\{x=k\}} \text{ a.e. } (\nu_1 + \nu_2) \end{aligned}$$

(f) $\mu = \text{Binom}(10, 0.5)$, $\nu = \text{Poisson}(1)$

$$\mu \ll \nu, \quad \forall A \in \mathcal{B}(\mathbb{R}), \mu(A) = \sum_{w \in A \cap \{0, \dots, 10\}} \binom{10}{w} 2^{-10}$$

$$= \int_A \binom{10}{w} \frac{e w!}{2^{10}} I_{\{0, \dots, 10\}}(w) d\nu(w)$$

$$\text{So } \frac{d\mu}{d\nu}(w) = \binom{10}{w} \frac{e w!}{2^{10}} I_{\{0, \dots, 10\}}(w) \text{ a.e. } (\nu).$$

Extra Problem

$(\mathcal{X}, \mathcal{F})$ = measurable space. μ, μ_1, μ_2, ν are σ -finite measures on $(\mathcal{X}, \mathcal{F})$

$$\mu_3 = \mu_1 + \mu_2$$

(a) μ_3 is a measure.

μ_3 is the sum of 2 σ -finite measures on $(\mathcal{X}, \mathcal{F})$

so μ_3 is also a measure on $(\mathcal{X}, \mathcal{F})$

(b) If $\mu_1 \ll \nu$, $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$.

By Proposition 6.1.2. (iv)

$$\mu_1 \ll \nu, \mu_2 \ll \nu \Rightarrow \mu_1 + \mu_2 \ll \nu.$$

(c) R-N. derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$.

$$\begin{aligned} \forall A \in \mathcal{F} \quad \mu_3(A) &= \mu_1(A) + \mu_2(A) = \int_A \frac{d\mu_1}{d\nu} d\nu + \int_A \frac{d\mu_2}{d\nu} d\nu \\ &= \int_A \left(\frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu} \right) d\nu = \int_A \frac{d\mu_3}{d\nu} d\nu. \end{aligned}$$

$$\Rightarrow \frac{d\mu_3}{d\nu} = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}.$$

(d) If $\mu \ll \nu$ and $\mu \perp \nu$ holds, show that μ is the zero measure.

$$\mu \perp \nu \Rightarrow \exists A \in \mathcal{F} \text{ s.t. } \mu(A) = 0, \nu(A^c) = 0.$$

$$\mu(A^c) = \int_{\mathcal{X}} I_{A^c} d\mu$$

$$\mu \ll \nu \Rightarrow \mu(A^c) = \int_{\mathcal{X}} I_{A^c} \frac{d\mu}{d\nu} d\nu \leq \int_{\mathcal{X}} I_{A^c} d\nu = \nu(A^c) = 0$$

$$\Rightarrow \mu = 0.$$

(e) $\mu_1 = \text{lognormal}(3, 0.5)$, $\mu_2 = \text{Geometric}(0.3)$, $\nu = \text{Poisson}(8)$

$$\mu_3 = \mu_{3a} + \mu_{3b} \text{ w.r.t. } \nu.$$

$$\mu_1 \ll$$